- Data Analysis LG — Multi-Agent Data Analysis with MCP & LangGraph
  - Table of Contents
  - Overview
  - Key Technologies
  - Project Structure
  - Architecture Overview
    - High-Level System Diagram
    - The Two-Process Model
    - Agent Team
  - How It Works
    - Startup Sequence
    - Agent Orchestration Flow
    - Tool Execution Cycle
  - MCP Tool Catalog
    - load_csv
    - get_dataframe_info
    - execute_python
    - list_output_files
    - write_markdown_report
  - Quick Start
    - Prerequisites
    - 1. Install Dependencies
    - 2. Set Up Environment
    - 3. Place Your Data
    - 4. Start the MCP Server
    - 5. Run the Analysis
    - 6. Check Results
  - Configuration
  - Further Reading

Data Analysis LG — Multi-Agent Data Analysis with MCP & LangGraph

> **A multi-agent system that performs automated exploratory data analysis on CSV datasets using LLM-powered agents, coordinated by LangGraph and communicating through the Model Context Protocol (MCP).**

Table of Contents

- Overview
- Key Technologies
- Project Structure
- Architecture Overview
  - High-Level System Diagram
  - The Two-Process Model
  - Agent Team
- How It Works
  - Startup Sequence
  - Agent Orchestration Flow
  - Tool Execution Cycle
- MCP Tool Catalog
- Quick Start
- Configuration
- Further Reading

Overview

This project demonstrates a **production-style multi-agent architecture** for automated data analysis. Instead of a single monolithic LLM call, the system decomposes data analysis into specialized roles — profiling, coding, analysis, and reporting — each handled by a dedicated agent.

The agents communicate through a **shared message state** managed by LangGraph, and they interact with data through **MCP tools** served over HTTP. The result is a system that can:

1. Load and profile any CSV dataset
2. Generate and execute Python code for EDA (charts, statistics, correlations)
3. Interpret the results with domain-aware reasoning
4. Produce a polished Markdown report with embedded chart references

```
User Question                          Output
┌──────────────┐                       ┌──────────────────────┐
│ "Analyze my  │  ──► Multi-Agent ──►  │ ✅ Markdown Report   │
```

```
│  sensor data"│        Pipeline        │ ✅ PNG Charts          │
└──────────────┘                        │ ✅ Statistical Tables  │
                                        └────────────────────────┘
```

Key Technologies

Technology	Role	Version
LangGraph	Multi-agent state machine & orchestration	≥ 0.2.0
MCP (Model Context Protocol)	Tool server/client protocol over HTTP	≥ 1.26.0
LangChain	LLM abstraction & tool interface	≥ 0.3.0
Groq	LLM inference (ultra-fast)	API
Pandas / **NumPy**	Data manipulation	≥ 2.0 / ≥ 1.24
Matplotlib / **Seaborn**	Visualization	≥ 3.7 / ≥ 0.12
Pydantic	Schema validation for tool arguments	≥ 2.0
Starlette + **Uvicorn**	HTTP server for MCP transport	latest

Project Structure

```
data_analysis_lg/
├── .env                    # GROQ_API_KEY (not committed)
├── main.py                 # 🚀 Entry point — wires MCP client + LangGraph
├── graph.py                # 🧠 Multi-agent LangGraph definition
├── client.py               # 🔌 MCP ↔ LangChain bridge
├── server.py               # 🖥️ MCP tool server (port 8002)
├── tools/
│   ├── __init__.py         # 📋 Central tool registry
│   ├── data_tools.py       # 📊 load_csv, get_dataframe_info
│   ├── python_executor.py  # 🐍 execute_python (sandboxed code runner)
│   └── report_tools.py     # 📝 list_output_files, write_markdown_report
├── csv/
│   └── example_data.csv    # 📁 Sample industrial sensor data (100K rows)
├── output/                 # 📂 Generated charts & reports land here
```

```
├── README.md              # ← You are here
├── ARCHITECTURE.md         # Detailed technical deep-dive
└── EXAMPLES.md             # Usage scenarios & sample outputs
```

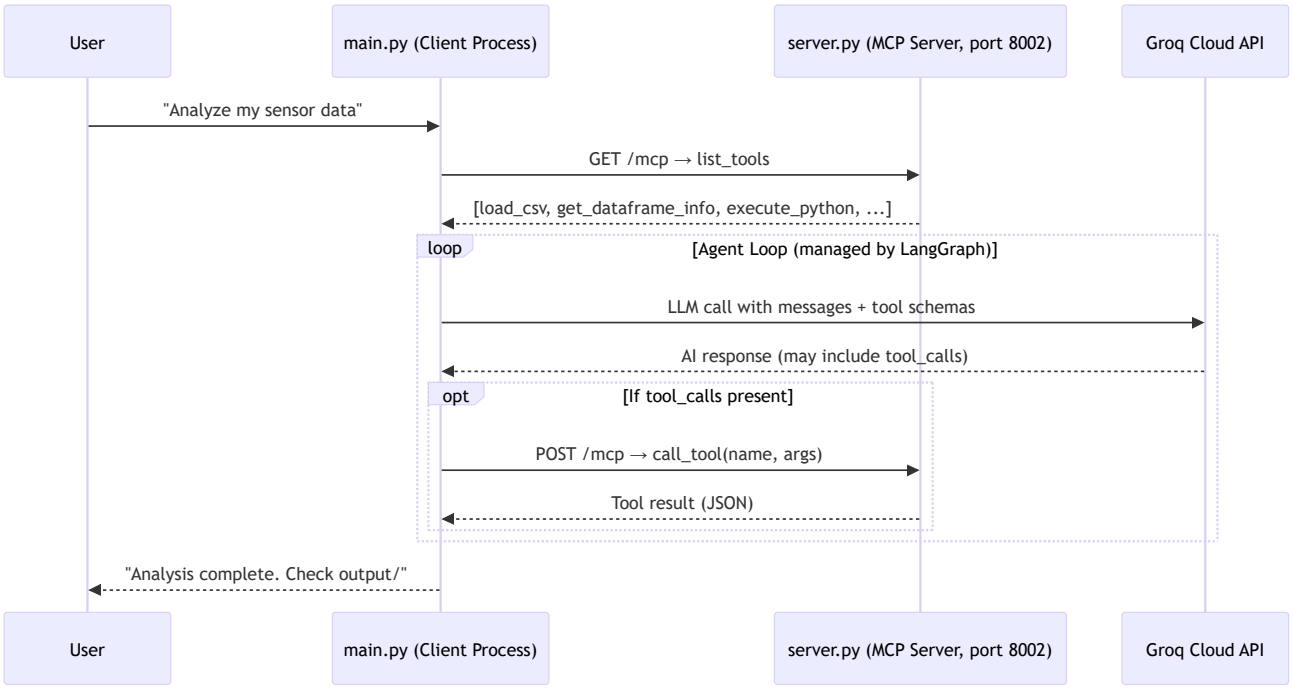
Architecture Overview

High-Level System Diagram

```
Parse error on line 2:
...ph TB    subgraph "Process 1 — MCP Tool...
----------------------^
Expecting 'SEMI', 'NEWLINE', 'SPACE', 'EOF', 'GRAPH', 'DIR',
'TAGEND', 'TAGSTART', 'UP', 'DOWN', 'subgraph', 'end', 'SQE', 'PE',
'-)', 'DIAMOND_STOP', 'MINUS', '--', 'ARROW_POINT', 'ARROW_CIRCLE',
'ARROW_CROSS', 'ARROW_OPEN', 'DOTTED_ARROW_POINT',
'DOTTED_ARROW_CIRCLE', 'DOTTED_ARROW_CROSS', 'DOTTED_ARROW_OPEN',
'==', 'THICK_ARROW_POINT', 'THICK_ARROW_CIRCLE',
'THICK_ARROW_CROSS', 'THICK_ARROW_OPEN', 'PIPE', 'STYLE',
'LINKSTYLE', 'CLASSDEF', 'CLASS', 'CLICK', 'DEFAULT', 'NUM', 'PCT',
'COMMA', 'ALPHA', 'COLON', 'BRKT', 'DOT', 'PUNCTUATION',
'UNICODE_TEXT', 'PLUS', 'EQUALS', 'MULT', got 'STR'
```

The Two-Process Model

The system runs as **two separate processes** that communicate over HTTP:



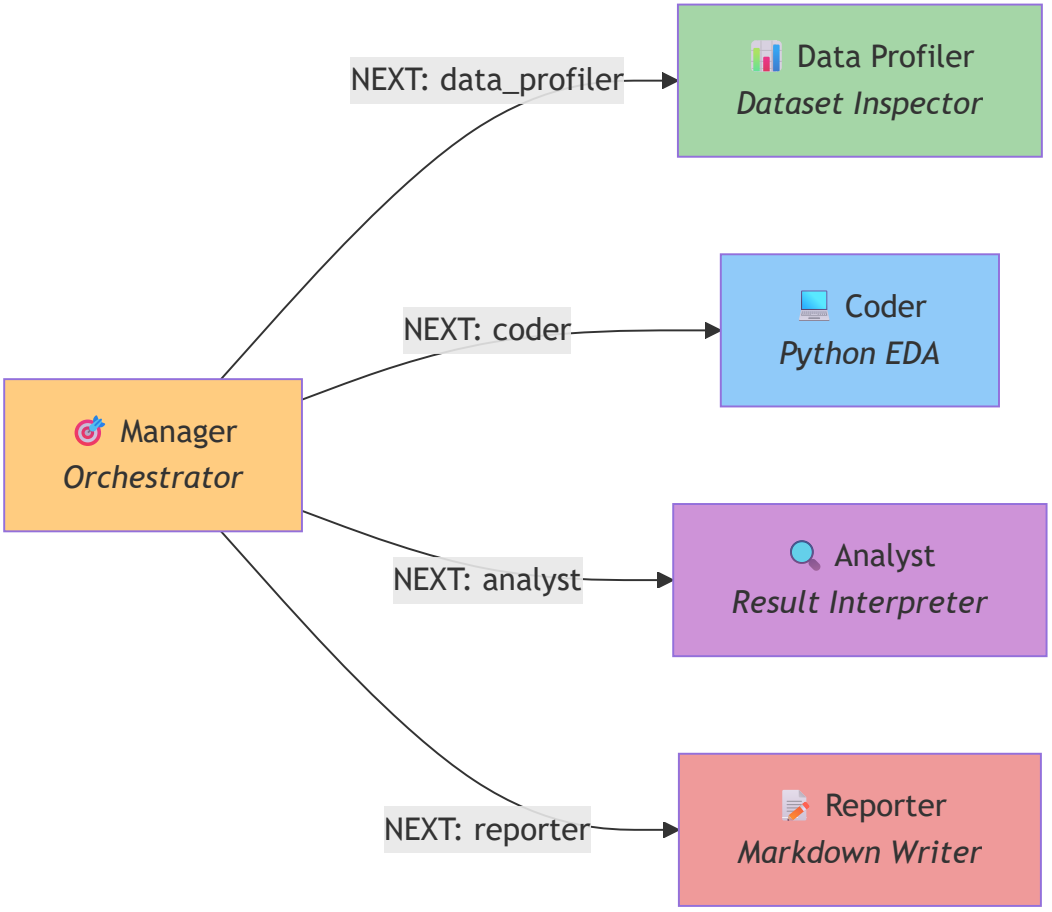
Why two processes?

Concern	Server (Process 1)	Client (Process 2)
Responsibility	Execute tools, manage data	Orchestrate agents, call LLM
State	Holds DataFrame in memory	Holds conversation messages
Protocol	MCP over Streamable HTTP	LangGraph state machine
Scaling	Could run on a different machine	Could run multiple analyses

This separation follows the **MCP specification** — tools are hosted as a service, and any MCP-compatible client can connect to them.

Agent Team

The system uses **5 specialized agents**, each with a focused role:

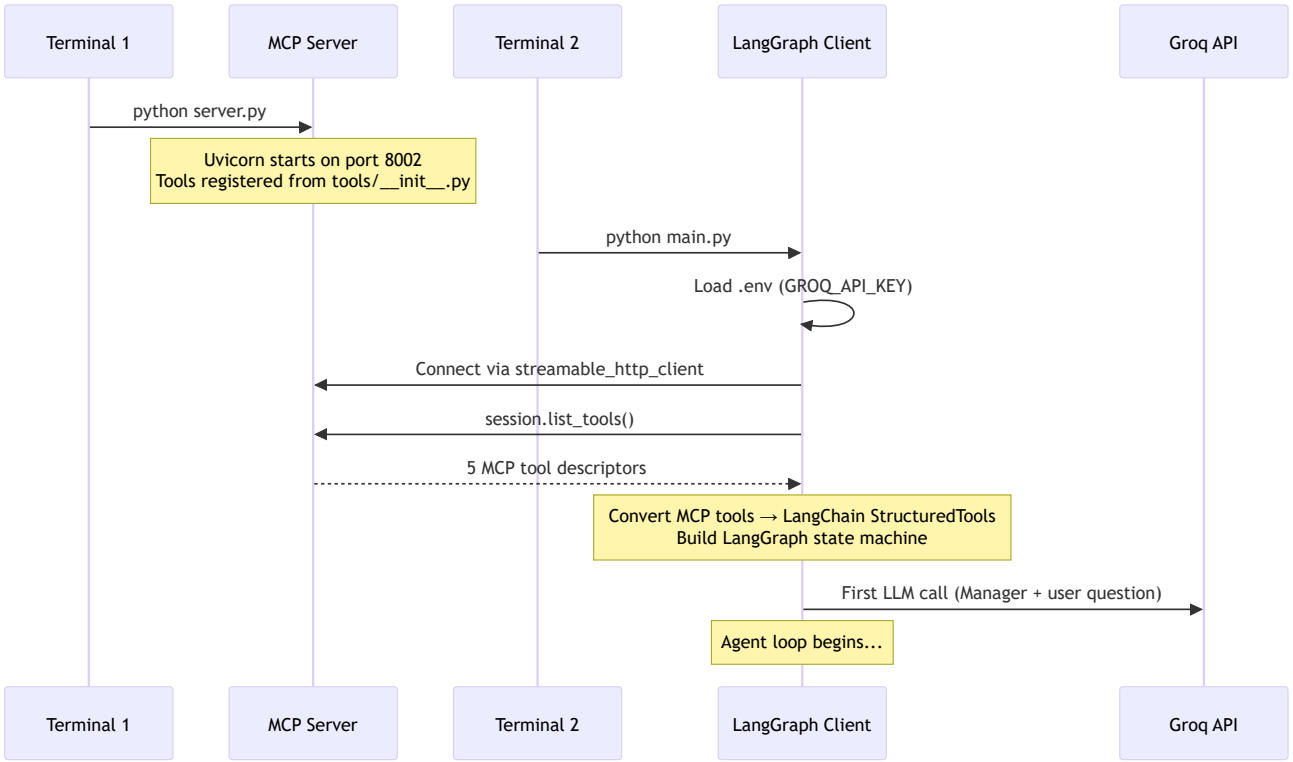


Agent	System Prompt Focus	Tools Used	Output
Manager	Route work to the right specialist	*None* (LLM only)	`NEXT: <agent>`

Agent	System Prompt Focus	Tools Used	Output
			directive
Data Profiler	Load CSV, inspect structure & stats	`load_csv`, `get_dataframe_info`	Dataset summary
Coder	Write & execute Python EDA code	`execute_python`	Charts (PNG) + printed stats
Analyst	Interpret charts, stats, patterns	`list_output_files`	Written analysis
Reporter	Assemble final Markdown report	`list_output_files`, `write_markdown_report`	`.md` report file

How It Works

Startup Sequence



Agent Orchestration Flow

This is the core loop that drives the entire analysis. The **Manager** acts as a router, reading the conversation history and deciding which specialist should act next:

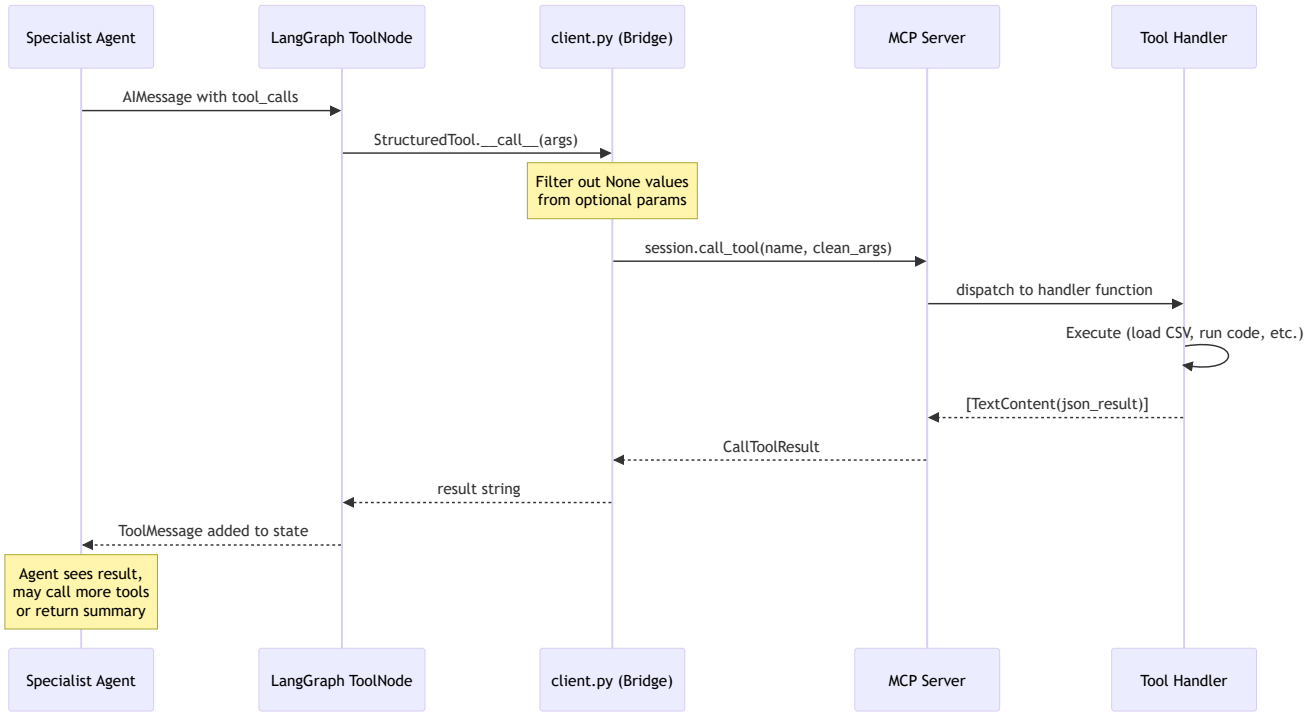
```
Parse error on line 1:
stateDiagram-v2
^
Expecting 'NEWLINE', 'SPACE', 'GRAPH', got 'ALPHA'
```

Key design decisions:

- Each specialist can call tools **multiple times** (the specialist ↔ tools loop continues until the agent responds without tool_calls)
- The Manager sees the **full compressed conversation** and decides the next step
- The `AnalysisState` carries both `messages` (conversation) and `next_agent` (routing signal)

Tool Execution Cycle

When a specialist agent decides to use a tool, the following happens:



MCP Tool Catalog

All tools follow the same three-part pattern:

```
1. Input Schema  (JSON Schema dict)    → Defines parameters
2. Tool Descriptor (mcp.types.Tool)    → Name + description for LLM
3. Async Handler  (async function)     → Executes the work
```

load_csv

> Load a CSV file into the server's in-memory DataFrame store.

Parameter	Type	Required	Description
`file_path`	string	✅	Path to CSV (relative to `csv/` folder or absolute)
`nrows`	integer	❌	Max rows to load (omit for all)

Returns: shape, column names, dtypes, first 2 rows as JSON.

---

get_dataframe_info

> Return detailed metadata about the loaded DataFrame.

Parameter	Type	Required	Description
`include_stats`	string	❌	Set to `"yes"` for descriptive statistics

Returns: shape, memory usage, numeric/non-numeric columns, null counts, optional stats summary.

---

execute_python

> Execute Python code with access to the loaded DataFrame and data science libraries.

Parameter	Type	Required	Description
`code`	string	✅	Python code to execute

Execution namespace includes: `df`, `pd`, `np`, `sns`, `plt`, `os`, `OUTPUT_DIR`

Returns: stdout capture, list of newly created files, any errors.

`list_output_files`

> List files in the `output/` directory.

Parameter	Type	Required	Description
`extension_filter`	string	❌	Filter by extension (e.g. `"png"`, `"md"`)

Returns: file count, list of `{name, size_kb}` objects.

`write_markdown_report`

> Write a Markdown report to the `output/` directory.

Parameter	Type	Required	Description
`filename`	string	✅	Report filename (e.g. `"analysis_report.md"`)
`content`	string	✅	Full Markdown content

Returns: status, file path, size, line count.

Quick Start

Prerequisites

- Python ≥ 3.11
- A Groq API key
- `uv` (recommended) or `pip`

1. Install Dependencies

```
# From the repository root (MCP_Playground/)
uv venv .venv
```

```
.venv\Scripts\activate          # Windows
uv pip install -e . --native-tls
```

2. Set Up Environment

Create `data_analysis_lg/.env`:

```
GROQ_API_KEY=gsk_your_key_here
```

3. Place Your Data

Put your CSV file in `data_analysis_lg/csv/`. The project ships with `example_data.csv` (100K rows of industrial sensor data).

4. Start the MCP Server

```
# Terminal 1
.venv\Scripts\activate
python data_analysis_lg/server.py
# Output: Server is running on http://localhost:8002/mcp
```

5. Run the Analysis

```
# Terminal 2
.venv\Scripts\activate
python data_analysis_lg/main.py
```

6. Check Results

```
data_analysis_lg/output/
├── analysis_report.md       # Markdown report
├── time_series.png          # Time series chart
├── correlation_heatmap.png  # Sensor correlation matrix
├── distribution_histograms.png
```

```
├── boxplot_outliers.png
└── ore_by_shift.png
```

Configuration

Setting	Location	Default	Description
`GROQ_API_KEY`	`.env`	—	Your Groq API key
`GROQ_MODEL`	`graph.py:37`	`openai/gpt-oss-120b`	LLM model name
`MAX_TOOL_OUTPUT_CHARS`	`graph.py:40`	`1500`	Max chars per tool response (token management)
`SERVER_URL`	`client.py:23`	`http://localhost:8002/mcp`	MCP server endpoint
`recursion_limit`	`main.py:67`	`50`	Max LangGraph iterations

Further Reading

- **ARCHITECTURE.md** — Deep technical dive: code walkthrough, state management, message compression, routing logic
- **EXAMPLES.md** — Example scenarios, sample outputs, how to write custom analysis requests